

Suggested Answers for 2018 Y6 H1 Chemistry Preliminary Examination

Paper 1:

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
C	A	D	C	C	D	D	B	D	D	C	D	C	C	B	A	D	C	C	B

21	22	23	24	25	26	27	28	29	30
A	B	D	A	A	A	B	C	B	B

Paper 2 Section A

1(a)(i)

$$K_c = \frac{[\text{CoCl}_4^{2-}]}{[\text{Co}(\text{H}_2\text{O})_6^{2+}][\text{Cl}^-]^4} \quad \text{units} = \text{mol}^{-4}\text{dm}^{12}$$

1(a)(ii)

	$\text{Co}(\text{H}_2\text{O})_6^{2+}(\text{aq})$	$+ 4\text{Cl}^-(\text{aq})$	$\rightleftharpoons$	$\text{CoCl}_4^{2-}(\text{aq})$	$+ 6\text{H}_2\text{O}(\text{l})$
I	1.80	4.00		0	-
C	-0.92	-4(0.92)		+0.92	-
E	0.88	0.32		0.92	-

$$\text{value of } K_c = \frac{(0.92)}{(0.88)(0.32)^4} = 99.7$$

1(a)(iii)

test	obsn	eqm pos	predominant species present
①	violet soln	near the centre	mixture of pink $\text{Co}(\text{H}_2\text{O})_6^{2+}$ ions and blue CoCl_4^{2-} ions
②	violet soln turns blue	right	CoCl_4^{2-} ions

1(a)(iv)

When test-tube 2 is placed in a beaker of hot water, the temperature of the solution increases. This causes the position of equilibrium to shift right, favouring the forward endothermic reaction.

This is in accordance to Le Chatelier's principle as the system attempts to counteract the increase in temperature by absorbing heat.

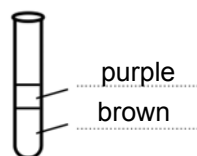
Hence with the forward reaction favoured, the concentration of pink $\text{Co}(\text{H}_2\text{O})_6^{2+}$ ions decreases, concentration of blue CoCl_4^{2-} ions increases, causing the solution to turn from violet to blue.

1(b)(i)

iodine in aqueous solution
iodine in hexane

brown
purple

1(b)(ii)



explanation

Hexane has a lower density than water. Since iodine dissolves in hexane to give purple colour, the purple layer is the upper layer.

1(b)(iii)

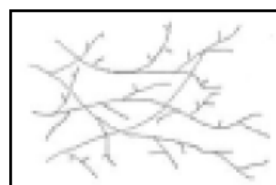
Iodine is non-polar and will form instantaneous dipole-induced dipole interactions with non-polar hexane molecules.

Iodine cannot form hydrogen bonds with polar water molecules.

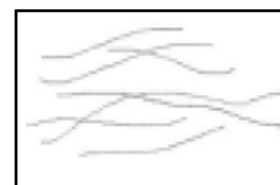
1(b)(iv)

Tri-iodide ion can form ion-dipole interaction with water. (which is stronger than the hydrogen bonding in water)

2(a)



LDPE



HDPE

HDPE is largely unbranched / linear and allows for more efficient/close packing of the polymer chains. Hence there is more mass packed into the same volume.

2(b)

Either one of these explanations and strategies

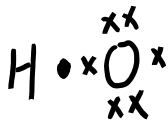
explanation	strategy
produced from crude oil, a non-renewable resource	use renewable resources
do not decompose/biodegrade	develop biodegradable polymers/
produce toxic gases on burning	recycle polymers

2(c)
elimination

3(a)

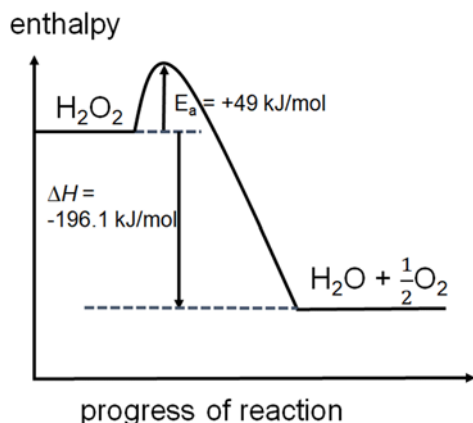
isotopic species	protons	neutrons	electrons	electronic configuration
$^{16}_8\text{O}$	8	8	8	$1s^2 2s^2 2p^4$
$^{18}_8\text{O}$	8	10	8	$1s^2 2s^2 2p^4$

3(b)



The free radical is highly reactive as it has a single unpaired electron.

3(c)(i)



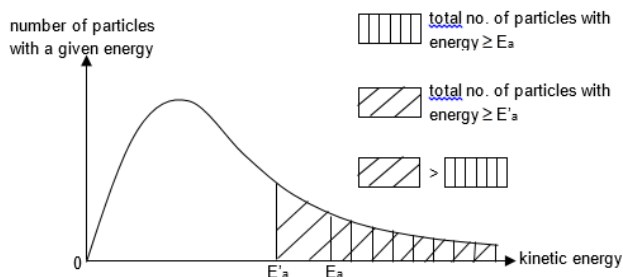
3(c)(ii)

The activation energy (E_a) of a reaction is the minimum amount of energy that the reactant particles must possess before their collisions can result in a reaction.

3(c)(iii)

In the catalysed reaction, since the activation energy is lower than that of the uncatalysed reaction, there is a greater proportion of reacting particles having energy greater than or equal to the activation energy than in the uncatalysed reaction.

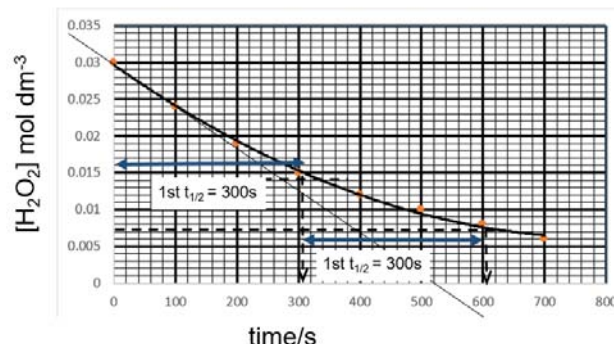
This is shown by the larger shaded area in the Maxwell-Boltzmann distribution curve.



Frequency of effective collision increases accordingly, and hence, the reaction rate increases.

Lower activation energy also results in the reaction having a larger rate constant, and hence, the reaction rate is increased.

3(d)(i)



3(d)(ii)

From the graph, $1^{\text{st}} t_{1/2} \approx 2^{\text{nd}} t_{1/2} = 300 \text{ s}$
Hence, the reaction is first order wrt H_2O_2 .

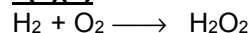
3(d)(iii)

Initial rate = $-\text{gradient of tangent to graph at time} = 0$
 $= 0.03 / 520 = 5.77 \times 10^{-5} \text{ mol dm}^{-3} \text{ s}^{-1}$

3(e)(i)

The product of oxidation is water which is clean/non-pollutants.

3(e)(ii)



3(e)(iii)

type of catalyst: heterogeneous catalyst
(as the catalyst and the reactants are in different phases)

For heterogeneous catalysis to occur, the reactant molecules need to be readily adsorbed onto the catalyst surface, i.e. the formation of weak bonds between the catalyst surface active sites and the reactant molecules.

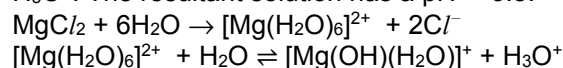
The adsorption of the reactant molecules at the catalyst surface increases the reaction rate because it

1. weakens the covalent bonds within the reactant molecules, thereby reducing the activation energy for the reaction.
2. increases the concentration of reactant molecules at the catalyst surface and allows the reactant molecules to come into close contact with proper orientation for reaction.

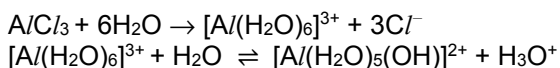
4(a)(i)

When MgCl_2 and AlCl_3 dissolves in water, the Mg^{2+} and Al^{3+} gets hydrated.

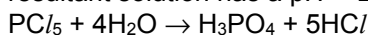
The hydrated Mg^{2+} undergoes slight hydrolysis to give H_3O^+ . The resultant solution has a $\text{pH} \approx 6.5$.



The hydrated Al^{3+} undergoes appreciable hydrolysis to give H_3O^+ . The resultant solution has a $\text{pH} \approx 3$.



PCl_5 hydrolyses in water to give H_3O^+ as shown. The resultant solution has a pH $\approx$ 2.

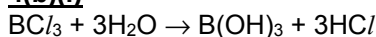


4(a)(ii)

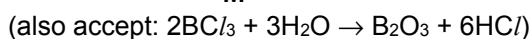
Ionic chlorides such as MgCl_2 dissolve in water to give an almost neutral (slightly acidic) solution

Covalent chlorides such as PCl_5 hydrolyse in water to give an acidic solution.

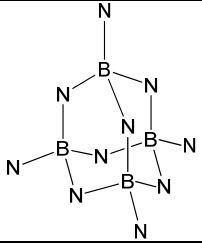
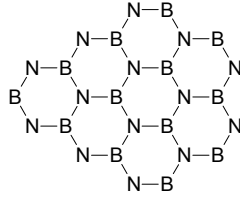
4(b)(i)



M



4(b)(ii)

	diamond-like form	graphite-like form
structure		
physical property	- high m.p./b.p. - or does not conduct electricity - or hard	- conducts electricity - or high m.p./b.p. - or soft

5(a)

alkene, 2° alcohol, 1° amine, chloroalkane (any 3)

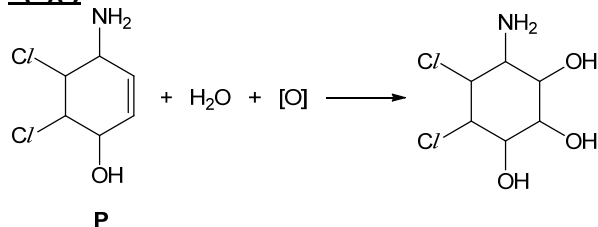
5(b)(i)

1640 – 1690 cm^{-1} (or 1475 – 1625 cm^{-1})

5(b)(ii)

700 – 800 cm^{-1}

5(c)(i)



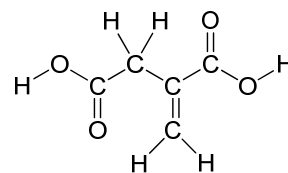
5(c)(ii)

step 2: ethanolic NaOH, heat
step 3: RCOOH , DCC

5(d)(i)

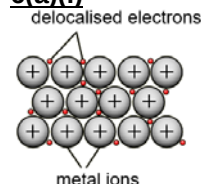
type of reaction: condensation
reagents and condns: (catalytic) conc H_2SO_4 , heat

5(d)(ii)



Section B

6(a)(i)



Iron exists as a giant metallic structure with strong electrostatic attraction / metallic bonds between positively charged iron ions and delocalised electrons.

6(a)(ii)

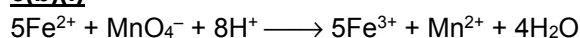
property: good electrical conductivity

explanation: presence of mobile charge carriers (delocalised electrons) in metallic structure.

property: high melting point

explanation: A large amount of energy is needed to break the strong electrostatic attraction between positively charged iron ions and delocalised electrons.

6(b)(i)



6(b)(ii)

In 25 cm^3 of solution

$$\text{Amt of } \text{MnO}_4^- = \frac{27.55}{1000} \times 0.005 = 1.3775 \times 10^{-4} \text{ mol}$$

$$\text{Amt of } \text{FeSO}_4 = \text{Amt of } \text{Fe}^{2+}$$

$$= 5 \times 1.3775 \times 10^{-4} = 6.8875 \times 10^{-4} \text{ mol}$$

In 250 cm^3 of solution

$$\text{Amt of } \text{FeSO}_4 = 10 \times 6.8875 \times 10^{-4} = 6.89 \times 10^{-3} \text{ mol}$$

6(b)(iii)

Mass of FeSO_4 in 5 tablets

$$= 6.8875 \times 10^{-3} \times 151.9 = 1.046 \text{ g}$$

$$\text{Mass of } \text{FeSO}_4 \text{ in 1 tablet} = 1.046 \div 5 = 0.209 \text{ g}$$

6(b)(iv)

As KMnO_4 will oxidise Cl^- to Cl_2 .

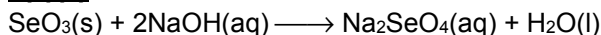
- volume of $\text{KMnO}_4(\text{aq})$ required for complete reaction is larger
- calculated amount of FeSO_4 in each tablet will be larger than actual
- calculated mass of FeSO_4 in each tablet will be larger than actual.

6(c)(i)

From its high melting point and position of Ge in the Periodic Table (diagonal relationship with Al), GeO_2 has a giant ionic structure with strong ionic bonds between Ge^{4+} and O^{2-} ions. (or GeO_2 is giant covalent)

From its low melting point and position of Se in the Periodic Table, SeO_3 has a simple covalent structure with weak instantaneous dipole-induced dipole interactions between molecules.

6(c)(ii)



6(d)(i)

Let x be the average oxidation number of C.

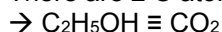
$$6x + 12(+1) + 6(-2) = 0 \rightarrow x = 0$$

6(d)(ii)

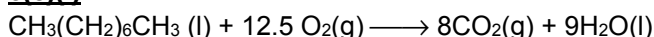
Average O.N. of C in ethanol = -2

O.N. of C in CO_2 = +4

There are 2 C atoms in ethanol and 1 C atom in CO_2



6(e)(i)



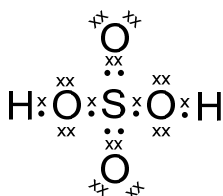
6(e)(ii)

$$\Delta H_{\text{rxn}}^\ominus = \sum n\Delta H_f^\ominus(\text{pds}) - \sum m\Delta H_f^\ominus(\text{rts})$$

$$= [8(-393.5) + 9(-285.8)] - [1(-249.7) + 12.5(0)]$$

$$= -5470 \text{ kJ mol}^{-1} \text{ (3 sf)}$$

7(a)(i)



It is tetrahedral around the S atom.

7(a)(ii)

$$\Delta H_f \text{ of HF} = \frac{1}{2} (-542) = -271 \text{ kJ mol}^{-1}$$

By Hess' Law,

$$\Delta H_f = \Delta H_3 + \Delta H_4 - \Delta H_1 - \Delta H_2$$

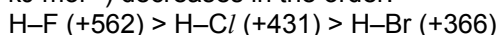
$$= 1220 + 814 - 542 - 1434 = +58 \text{ kJ mol}^{-1}$$

7(b)(i)

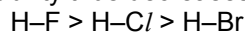
Thermal stability of the hydrogen halides (HX) is related to the H-X bond strength.

More endothermic bond dissociation energy suggests that the H-X bond is stronger and that HX is more thermally stable.

Since bond dissociation energy of H-X bond (in kJ mol^{-1}) decreases in the order:



thermal stability thus decreases in the order:



7(b)(ii)

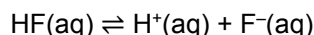
Since HCl is a strong acid,

$$[\text{H}^+] = [\text{HCl}] = 0.050 \text{ mol dm}^{-3}$$

$$\text{pH} = -\lg(0.050) = 1.30$$

7(b)(iii)

HF is partially dissociated (or ionised) in aqueous solution.

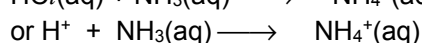
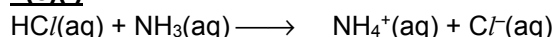


7(b)(iv)

(1) The K_a of the acid remains constant when it is diluted as it is a constant at constant temperature.

(2) Dilution will result in a lower $[\text{H}^+(\text{aq})]$. Since $\text{pH} = -\lg[\text{H}^+]$, pH should increase with dilution.

7(c)(i)



7(c)(ii)

methyl orange

yellow to orange

7(c)(iii)

$$\text{Amt of HCl} = (22.00/1000)(0.050) = 1.10 \times 10^{-3} \text{ mol}$$

$$[\text{NH}_3(\text{aq})] = 1.10 \times 10^{-3} / (25.0/1000)$$

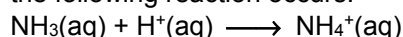
$$= 0.0440 \text{ mol dm}^{-3}$$

7(c)(iv)

NH_3 and NH_4^+ is a buffer solution which resists changes in pH upon the addition of a small amount of acid or base.

Or it contains a large reservoir of weak base, NH_3 , and its conjugate acid, NH_4^+ .

When a small amount of H^+ ions is added to the solution, the following reaction occurs:



The presence of a large reservoir of unionised NH_3 molecules in the solution ensures that nearly all the added H^+ ions are removed.

Hence $[\text{H}^+]$ in the solution changes very little and the pH is kept approximately constant.